

Investment Risk Assessment through Machine Learning: An Extension of the Automated Investment Questionnaire Tool

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1. EXECUTIVE SUMMARY

This project presents a technical enhancement of the Automated Investment Questionnaire Tool. Initially designed to digitize client intake, improving processing time by 50% where this extension introduces a predictive layer. By utilizing an XGBoost classifier, the tool provides a data driven benchmark to validate subjective risk assessments. The expected impact includes a significant reduction in human advisory bias and a robust audit trail for regulatory compliance.

2. PROBLEM STATEMENT

Traditional financial advisory relies heavily on "Risk Tolerance" (subjective feeling) rather than "Risk Capacity" (objective ability to afford loss). This reliance on human intuition introduces cognitive bias into asset allocation. There is a critical need for an analytical support facility that uses empirical peer data to provide a statistically significant "reality check" against client-reported preferences.

3. BUSINESS OBJECTIVES

The primary business goals are as follows:

Decision Accuracy: Use historical patterns to validate or challenge client reported risk levels.

Operational Efficiency: Build upon previous digital gains that reduced paper-based application processing time by 50%.

Regulatory Alignment: Provide a transparent, data-backed rationale for asset allocation to meet "Know Your Client" (KYC) standards.

4. DATA COLLECTION AND SOURCES

The predictive engine utilizes the **USA Survey of Consumer Finances (SCF)**, a triennial cross-sectional study conducted by the Federal Reserve [1].

Data Type: Tabular, cross-sectional financial survey data.

Dimensions: The raw dataset consists of 22,795 rows and 357 columns.

Acquisition: Data was extracted via Federal Reserve public repositories, focusing on household balance sheets, income, and demographics.

5. DATA EXPLORATION AND PREPROCESSING

Initial Data Analysis (IDA) identified high dimensionality and class imbalance.

Cleaning: Resolved entries with duplicate ID fields and managed missing values in pension and asset categories.

Normalization: Categorical encoding was applied to occupation and family structure data. Outlier detection was performed on extreme net-worth values to prevent model skewing.

6. FEATURE ENGINEERING

The initial 357 columns were filtered to isolate high-impact features.

- **AGECL:** Raw age was binned into six standard industry age clusters (1–6).

- Net Worth: Engineered by calculating the delta between Total Assets and Total Debt.
- Solvency Ratios: Selection of debt-to-income (DTI) and equity-to-asset metrics to serve as objective indicators of financial capacity.

7. MODEL SELECTION AND TRAINING

The project performed a comparative analysis between **Random Forest** and **XGBoost** [2].

Rationale: Random Forest demonstrated susceptibility to high correlation among financial variables, leading to overfitting (1.0 Accuracy). XGBoost was selected for its superior ability to handle tabular financial data and handle biased distributions via gradient boosting.

Training: Conducted using an 80/20 train-test split within a Python-based development environment.

8. MODEL EVALUATION

1st test findings: The model overall has a 94% score. However, the precision score for aggressive risk profile is only 0.39 or 40%. The F1 Score also is low to 51%. Target: Raise precision to higher percentage.

2nd test findings: The model was subjected to manual weights, where the "aggressive group" with the lowest number of records were given a weight of 5 to balance the sensitivity of the model to that group. The model was then run and evaluated. The model achieved a Weighted Average F1-score of 0.95 and a higher precision score for the aggressive group, that we wanted.

Table 1. XGBoost Classification Performance

Class	Precision	Recall	F1-Score	Support
Conservative	0.93	0.96	0.95	385
Moderate	0.96	0.95	0.95	519
Aggressive	0.89	0.53	0.67	15

9. MODEL DEPLOYMENT

The tool is deployed as a modular streamlit application on Streamlit Cloud. It was also organized into a project folder then upload to github containing the base application, models, csv files, and the requirements.txt.

Infrastructure: GitHub-hosted repository utilizing a Linux-based cloud environment.

Architecture: Model-loading logic is centralized in src/model_load.py, accessed via an @st.cache_resource wrapper to optimize performance.

10. MONITORING AND MAINTENANCE

Logging: A local invq_v2.log tracks prediction events and system health.

Audit **Trail:** A persistent client_recommendations.csv records side-by-side comparisons of subjective and objective results.

Update Strategy: The model will be retrained upon the release of new triennial SCF survey results to detect shifts in consumer financial behavior.

11. ETHICAL CONSIDERATIONS AND BIAS MITIGATION

Mitigation: Adjusted XGBoost weights to ensure minority risk profiles (Aggressive) were not ignored by the model.

Transparency: The tool explicitly presents the AI recommendation as a "Peer-based reality check," maintaining a "Human-in-the-loop" approach where the advisor makes the final decision.

12. PROJECT TIMELINE AND MILESTONES

Phase 1: Understanding the business problem and identifying the approach.

Phase 2: Perform EDA on the dataset.

Phase 3: Perform feature engineering, model selection and evaluation. Saved and loaded the model into the streamlit application as an additional functionality. Aligned input fields and conformed the fields to the data attributes.

13. RISKS AND CHALLENGES

Overfitting: Initially encountered with Random Forest; mitigated by switching to XGBoost and tuning regularization.

Pathing: Resolved file path issues on Streamlit Cloud by utilizing `os.path` functions to locate the model within the singular `model/` folder.

14. APPENDICES

- [1] Board of Governors of the Federal Reserve System, "Survey of Consumer Finances (SCF)," Washington, DC, 2024. <https://www.federalreserve.gov/econres/scfindex.htm>
- [2] J. Brownlee, "XGBoost for Classification," *Machine Learning Mastery*, 2024. <https://machinelearningmastery.com/xgboost-with-python/>